

Triple Data Encryption Standard (3DES)

Aim

To implement the **Triple Data Encryption Standard (3DES)** algorithm for encrypting and decrypting plaintext using **three DES operations** and secret keys, thereby providing stronger security than DES and 2DES.

Algorithm

Encryption Algorithm (EDE Mode)

Step 1: Start the program.

Step 2: Read the plaintext from the user.

Step 3: Read the first secret key (K1).

Step 4: Read the second secret key (K2).

Step 5: Read the third secret key (K3).

Step 6: Encrypt the plaintext using the first key (K1).

$$C_1 = E_{K1}(P)$$

Step 7: Decrypt the intermediate ciphertext using the second key (K2).

$$C_2 = D_{K2}(C_1)$$

Step 8: Encrypt the result again using the third key (K3).

$$C = E_{K3}(C_2)$$

Step 9: Display the final ciphertext.

Step 10: For decryption:

- Decrypt using **K3**.
- Encrypt using **K2**.
- Decrypt using **K1**.

$$P = D_{K1} \left(E_{K2} (D_{K3} (C)) \right)$$

Step 11: Display the original plaintext.

Step 12: Stop the program.

C Program (Simplified 3DES Demonstration)

```

#include <stdio.h>

#include <string.h>

int main()
{
    char plaintext[100], key1[100], key2[100], key3[100];
    char step1[100], step2[100], cipher[100], decrypted[100];
    int i, len;

    printf("Enter Plaintext: ");
    scanf("%s", plaintext);
    printf("Enter Key1: ");
    scanf("%s", key1);
    printf("Enter Key2: ");
    scanf("%s", key2);
    printf("Enter Key3: ");
    scanf("%s", key3);

    len = strlen(plaintext);

    // Encryption with Key1
    for(i = 0; i < len; i++)
        step1[i] = plaintext[i] ^ key1[i % strlen(key1)];

    // Decryption with Key2 (XOR demonstration)
    for(i = 0; i < len; i++)
        step2[i] = step1[i] ^ key2[i % strlen(key2)];

    // Encryption with Key3
    for(i = 0; i < len; i++)
        cipher[i] = step2[i] ^ key3[i % strlen(key3)];

    printf("\nEncrypted Cipher Text: ");
    for(i = 0; i < len; i++)
        printf("%02X", (unsigned char)cipher[i]);

    // Decryption with Key3
    for(i = 0; i < len; i++)

```

```

        step2[i] = cipher[i] ^ key3[i % strlen(key3)];

// Encryption with Key2
for(i = 0; i < len; i++)

    step1[i] = step2[i] ^ key2[i % strlen(key2)];

// Decryption with Key1
for(i = 0; i < len; i++)

    decrypted[i] = step1[i] ^ key1[i % strlen(key1)]

decrypted[len] = '\0';

printf("\nDecrypted Plaintext: %s\n", decrypted);

return 0;

}

```

OUTPUT :

The screenshot shows an IDE window titled "D:\CSA5112 - Cryptography\LAB PROGRAMS\3DES.cpp" with the following code:

```

23 // Encryption with Key1
24 for(i = 0; i < len; i++)
25     step1[i] = plaintext[i] ^ key1[i % strlen(key1)];
26
27 // Decryption with Key2 (XOR demonstration)
28 for(i = 0; i < len; i++)
29     step2[i] = step1[i] ^ key2[i % strlen(key2)];
30
31 // Encryption with Key3
32 for(i = 0; i < len; i++)
33     cipher[i] = step2[i] ^ key3[i % strlen(key3)];
34
35 printf("\nEncrypted Cipher Text: ");
36 for(i = 0; i < len; i++)
37     printf("%02X", (unsigned char)cipher[i]);
38
39 // Decryption with Key3
40 for(i = 0; i < len; i++)
41     step2[i] = cipher[i] ^ key3[i % strlen(key3)];
42
43 // Encryption with Key2
44 for(i = 0; i < len; i++)
45     step1[i] = step2[i] ^ key2[i % strlen(key2)];
46
47 // Decryption with Key1
48 for(i = 0; i < len; i++)
49     decrypted[i] = step1[i] ^ key1[i % strlen(key1)];
50
51 decrypted[len] = '\0';
52 printf("\nDecrypted Plaintext: %s\n", decrypted);
53
54 return 0;
55
56
57

```

The console output shows the program's execution:

```

Enter Plaintext: HELLO
Enter Key1: KEY1
Enter Key2: KEY2
Enter Key3: KEY3

Encrypted Cipher Text: 0300157C04
Decrypted Plaintext: HELLO

-----
Process exited after 18.07 seconds with return value 0
Press any key to continue . . .

```

The IDE interface includes a menu bar (File, Edit, Search, View, Project, Execute, Tools, AStyle, Window, Help), a toolbar, a project explorer, a compiler window showing 0 errors and 0 warnings, and a status bar at the bottom indicating line 57, column 7, and parsing time.